

Bigram Models

- Condition on previous single word
- Generative process:
 - pick START, pick a word conditioned on previous one, repeat until STOP
- The model can be factored thusly:

$$p(x_1 \dots x_n) = \prod_{i=1}^n p(x_i | x_{i-1})$$

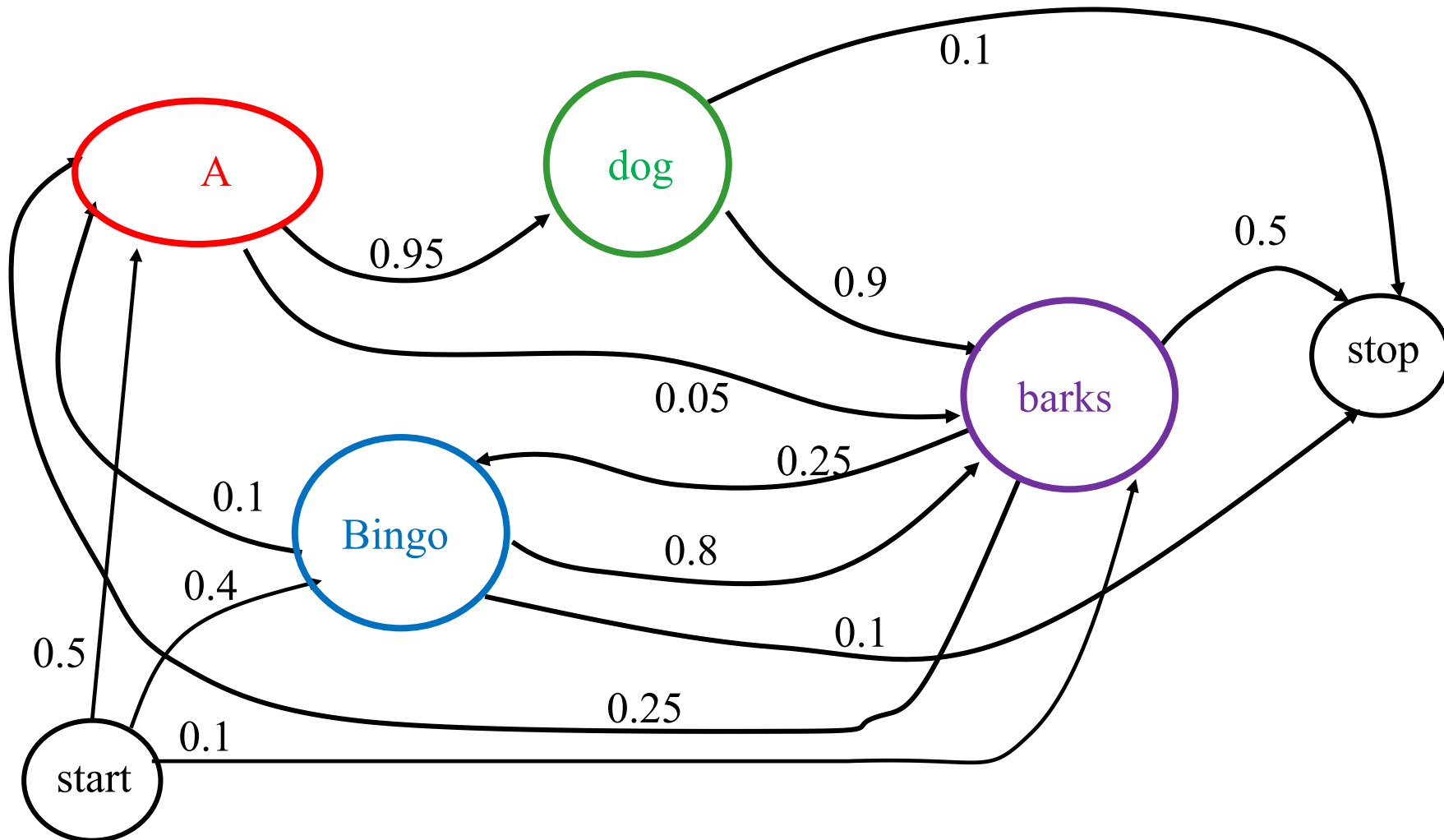
- This is equivalent to assuming the words of the sentence are (homogeneous, 1st order) Markov chains, namely that:

$$p(x_i | x_{i-1} \dots x_0) = p(x_i | x_{i-1})$$

Markov Model / Markov Chain

- A finite state automaton with probabilistic state transitions
- Makes Markov assumption that next state only depends on the current state and independent of previous history **given the current state**

Sample Markov Model for LM



Formal Representation of Markov LMs

- The parameters of a Markov LM can be represented as a stochastic, positive-valued matrix (*transition matrix*)
 - Sometimes the distribution of the first symbol is represented separately, but we can think about the sequences as always starting with n instances of a special START symbol

$$A_{ij} = P(x_m = w_j | x_{m-1} = w_i)$$

$$w_i, w_j \in WORDS \cup \{START, STOP\}$$

- Example:

	START	FUNNY	LITTLE	THINGS	STOP
START	0	0.5	0.5	0	0
FUNNY	0	0.8	0.1	0.1	0
LITTLE	0	0.1	0.1	0.8	0
THINGS	0	0	0	0.1	0.9
STOP	0	0	0	0	1